

Janhavi Sharma

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Education

Rutgers University, New Brunswick, NJ

United States

Master of Science in Data Science (GPA 3.9/4.0)

May 2026

- **Relevant Coursework:** Regression & Time Series Analysis, Probability & Statistical Inference, Financial Data Mining, Statistical Modeling & Computing, Neural Networks and Reinforcement Learning, Natural Language Processing, Financial Risk Management.

Thapar Institute of Engineering and Technology

India

Bachelors of Engineering in Computer Engineering (GPA 3.8/4.0)

June 2023

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Machine Learning, Deep Learning.

Skills & Certifications

- **Programming Languages:** C++, Python (NumPy, Pandas, Scikit-Learn, PyTorch), R, SQL, (PostgreSQL, Oracle), Matlab, Unix, AWS, Azure
- **ML Algorithms & Tools:** TensorFlow, Keras, Transformers, Linux, PySpark, Hadoop, Hive, Tableau, Power BI, Apache, Spark, Databricks, Airflow
- **Certifications:** Top 5,000 Global Rank- *HackerRank*, **Bloomberg** Financial Fundamentals & Market Concepts, **NVIDIA** Accelerated DL.

Work Experience

Rutgers University, SAS IT & Data Analytics Office

New Jersey, United States

Data Scientist Intern

May 2025 - August 2025

- **Developed** a centralized housing database across 4 campuses using AWS S3 & SQL, consolidating fragmented student housing records.
- **Built** and automated SQL ETL pipelines for ingestion, cleaning, and validation, ensuring reliable and error-free data delivery.
- **Applied** clustering (DBSCAN, K-Means) and Random Forests to detect trends in housing preferences, gender & major selections.
- **Created** dashboards in Tableau to visualize trends and present to university stakeholders for strategic planning and resource optimization.

ION Trading

Delhi, India

Software Developer

January 2023 - March 2024

- **Developed** trade gateway pipelines in C++ to process real-time market data by applying object-oriented programming principles.
- **Migrated** historical and streaming trade data to AWS(S3, Lambda) using Python, achieving a 40% reduction in latency.
- **Acquired** expertise in financial instruments, including spreads, equities, bonds, commodities, and risk management strategies.
- **Leveraged** Python and SQL to manage databases and extract actionable insights from market and client trade data.
- **Built** a python-based anomaly detection tool to identify and analyze trends in client trade logs, enhancing decision-making.
- **Acted** as primary point of contact for high-priority client issues, resolving market data inconsistencies and pipeline failures.

Medly.

New York, United States

Data Analyst Intern

June 2022 - September 2022

- **Delivered** a 25% improvement in product solutions by analyzing CRM and user data with Python and SQL.
- **Designed** Tableau and Power BI dashboards to track sales pipelines and user engagement metrics.
- **Developed** lead scoring and customer segmentation models using Probabilistic GRUs (to model user behaviour sequences), regularized Multi-Layer Perceptrons (MLPs), and Variational Autoencoders (VAE), to improve targeting and customer retention.
- **Conducted** A/B testing and statistical analysis with SciPy and statsmodels, optimizing new feature rollouts.

Projects

Time Series Forecasting and Model Evaluation for Air Quality Prediction

- **Built** forecasting models- AR, MA, ARIMA, SARIMA, XGBoost, LightGBM, Prophet, LSTM to predict air pollution levels.
- **Optimized** models for daily predictions, focusing on capturing trends, seasonality, and anomalies.
- **Evaluated** models using MAE, RMSE, MAPE, and R^2 metrics, with XGBoost & LightGBM delivering the highest accuracy.

Performance Benchmarking of YOLO and SSD Models for Object Detection

- **Investigated** one-shot detection algorithms, including YOLO (versions 6–9) and SSD, for automated visual recognition tasks.
- **Benchmarked** model performance on COCO and ImageNet datasets, assessing accuracy using transfer learning techniques.
- **Evaluated** efficiency using Mean Average Precision (mAP) and accuracy metrics.

Image Scribe (Computer Vision + NLP)

- **Implemented** an image-to-text model to generate natural language captions.
- **Designed** an encoder-decoder architecture using VGG16 CNN for feature extraction and LSTM for sentence generation.
- **Trained** the model on Flickr8K dataset, achieving high accuracy with BLEU score, and integrated Text-to-Speech functionality.

LunarLander Reinforcement Learning Agent

- **Designed** and trained an RL agent for the LunarLander environment using Gymnasium.
- **Implemented** and compared SARSA, Q-learning, and Deep Q-Learning algorithms to optimise agent performance.
- **Optimized** learning using a replay buffer and epsilon-greedy strategy for effective exploration-exploitation trade-off.

Open Eye Detector Real-time drowsiness detection (using CNNs with OpenCV & Tensorflow and, Haar cascades)

Pet Activity Detection: (Behaviour classification using K-means, Gaussian HMM, and LSTM)